# DPM Expression Language (DPM-XL) Documentation

**None**

*EBA and EIOPA*

*None*

# Table of contents

1. DPM Expression Language (DPM-XL) Documentation 3
1.1 What is DPM-XL? 3
1.2 Key Features 3
1.3 Quick Start 3
1.4 Documentation Structure 3
1.5 Who Should Use This Documentation? 4
1.6 Getting Help 4
2. Information Model 5
2.1 Introduction 5
2.2 2 Information Model 7
2.3 3 DPM-XL Data Selection 15
2.4 4 Null in DPM Expression Language 16
2.5 5 DPM-ML Metamodel 17
3. Operators 22
3.1 1 Introduction 22
3.2 2 Common General Behaviours for the Operators 23
3.3 3 Selection Operators 26
3.4 4 Numeric Operators 29
3.5 5 Comparison Operators 38
3.6 6 Logical Operators 45
3.7 7 Aggregate Operators 48
3.8 8 Conditional Operators 53
3.9 9 String Operators 57
3.10 10 Time Operators 59
3.11 11 Clause Operators 61
4. Grammar 65
4.1 Introduction 65

# 1. DPM Expression Language (DPM-XL) Documentation

Welcome to the comprehensive documentation for the **Data Point Model Expression Language (DPM-XL)**, a formal language for expressing data validation and calculation requirements in regulatory reporting.

## 1.1 What is DPM-XL?

The DPM Expression Language is a formal, business-user-friendly language that enables the specification of:

- **Data validation rules** for regulatory reporting
- **Calculation formulas** for derived data points
- **Data transformation operations** across reporting tables
- **Cross-module validations** and consistency checks

DPM-XL serves as the foundation for automated validation systems used by regulatory authorities such as EBA and EIOPA.

## 1.2 Key Features

### 1.2.1 🎯 Business-Oriented

Designed for business users who understand regulatory requirements but may not be programmers.

### 1.2.2 ⚡ Formally Specified

Fully formalized syntax and semantics enable automated compilation and execution.

### 1.2.3 🔄 Dual Representation

- **DPM-XL**: Human-readable expression language
- **DPM-ML**: Machine-readable metamodel representation

### 1.2.4 📊 Rich Data Model

Supports complex data structures including scalars, sets, and multi-dimensional recordsets.

### 1.2.5 🛡️ Robust

Comprehensive null handling, type system, and error management.

## 1.3 Quick Start

Here's a simple DPM-XL expression:

```
{F_01.01, r0010, c0010} = {F_01.01, r0020, c0010} + {F_01.01, r0030, c0010}
```

This expression validates that the value in table F_01.01, row 0010, column 0010 equals the sum of rows 0020 and 0030 in the same column.

## 1.4 Documentation Structure

This documentation is organized to take you from basic concepts to advanced implementation:

### 1.4.1 🏗️ Information Model

Understand the core data structures: operations, artifacts (scalars, sets, recordsets), and data types.

### 1.4.2 ⚙️ Operators

Comprehensive reference for all operators, organized by category: - Selection, Numeric, Comparison, Logical - Aggregate, Conditional, String, Time, Clause

### 1.4.3 📖 Grammar

Quick reference materials, grammar specifications, and operator summaries.

## 1.5 Who Should Use This Documentation?

### 1.5.1 **Regulatory Specialists**

Writing validation rules and calculations for regulatory frameworks.

### 1.5.2 **Software Developers**

Building engines and tools that execute DPM-XL expressions.

### 1.5.3 **Implementation Teams**

Integrating DPM-XL into reporting systems and validation pipelines.

### 1.5.4 **Standard Bodies**

Understanding the formal specification for governance and evolution.

## 1.6 Getting Help

This documentation aims to be comprehensive and self-contained. Each section includes:

- Clear explanations of concepts
- Formal specifications where needed
- Practical examples
- Edge case handling

For the most current information and updates, please refer to the official DPM standards documentation.

---

**Navigation Tip**

Use the navigation menu on the left to explore different sections. Each section builds upon previous concepts, so we recommend following the order for first-time readers.

# 2. Information Model

## 2.1 Introduction

This is a technical document specifying the Information Model, the *Data Types* and other technical characteristics of the DPM *Operations*.

This documentation is mainly addressed to technical audiences with interest in developing engines or in having a better technical understanding of the DPM *Operations*. The document is not addressed to normal business users that write calculations, although it can be useful for those users with basic knowledge of information models and programming languages.

In any case, this document is complemented by the document describing the *Operators* used for the *Operations*, which should be the main reference for users writing calculations.

### 2.1.1 Why

EBA and EIOPA have been using for some years a semi-formal expression language for business users to write and share their data validation requirements.

The fact that is semi-formal allows automating some aspects for translating the expression into another language (e.g., XBRL Formula Link base). But because it is not completely formal, it is, by definition, not possible to create a compiler that fully automates the translation to another language or an interpreter for it.

This document, and its complementary document on the *Operators*, aims to provide this required formalization so that DPM *Operations* become fully formalized, therefore enabling the automation regarding the calculations written by EBA and EIOPA.

### 2.1.2 DPM-XL and DPM-ML

The DPM *Operations* have two representations: As expressions, with the DPM eXpression Language (DPM-XL) and as a structured representation in the database following a metamodel (DPM Metamodel Language or DPM-ML).

The reason for having **DPM-XL** is that it is the language business users write and can understand. Because it is the input of all the process, it is important to keep this language formal, so that it can be translated and executed. But, at the same time, it must serve for communication, so it has to be business users friendly, implying, among others, that the *Operands* of the language are referred to by using rendering artifacts (tables, row, columns and sheets).

A key requirement for the DPM-XL has been that the language that is currently used should change as little as possible. So, in practice, the formalization of the language has been a reverse-engineering process, where the starting point where the validations published by the EBA and EIOPA.

The main reasons for having **DPM-ML** are that it is providing a structured version of the calculations, with no need to parse expressions, and that it is referring to *Variables*, which are stable over time, because they represent business concepts, instead of rendering objects, which change often over time and do not hold any business meaning, but just a representation.

The DPM-ML is automatically derived from the DPM-XL and, eventually, can be also derived from languages other than the DPM-XL. DPM-ML relations to languages other than DPM-XL is out of the scope for this documentation. Instead, the documentation provides details on how *Operations* are represented both in DPM-XL and DPM-ML. The DPM-XL and DPM-ML share the same *Operators* and the same information model.

### 2.1.3 How

The formalization of the language has three pillars:

1. **Information model**: Specifies the artifacts that the language is using.
2. **Grammar**: Technical definition of the syntax of the language. Allows developers to build parsers for the language. It is provided in a separate file following the EBNF (Extended Backus-Naur form) notation (Only applicable to DPM-XL).
3. **Semantic specification**: Semantics for all *Operators* of the language, specifying formally their constraints and behaviour.

This document deals with the information model, while the grammar and semantic specification are provided as separate documents.

The Validation and Transformation Language specification has been used as an inspiration for this specification.

# 2.2 2 Information Model

## 2.2.1 2.1 Operations

classDiagram class Script class Operation class Result class Operand class Operator class LanguageArtifact class Output Class Input Script *-- Operation Operation --> "1..1" Result : Produces Operation --> "0..n" Operand : Acts on Operation --> "0..n" Operator: Uses Result --|> LanguageArtifact Operand --|> LanguageArtifact Operator --> "1..1" Output : Produces Operator --> "0..n" Input : Uses Output --> Input Operand --> Input style Script fill:#0098f6 style Operation fill:#0098f6 style Result fill:#0098f6 style Operand fill:#0098f6 style Operator fill:#0098f6 style LanguageArtifact fill:#04f286 style Input fill:#0098f6 style Output fill:#0098f6

The DPM *Operations* serve to write *Scripts*, which are computer programs that constitute a run-time and are composed of *Operations*.

*Operations* are expressions that use input *Operands* and/or *Operators* to produce a result.

Expressions are finite combinations of symbols that are well-formed according to the syntactical rules of the language. Expressions compose some *Operands* in a certain order by means of the *Operators* of the language, to obtain the desired result. The symbols of the expression designate *Operators*, *Operands*, and the order of application of the *Operators*.

*Operators* specify a type of operation to be performed on some input *Operands* (exceptionally, there may be *Operators* that do not take *Operands* as input, e.g., an *Operator* to get the current time) to generate an output. The output produced by one *Operator* may be used as input for another *Operator* (i.e., *Operators* can be nested).

*Operands* are specific artifacts from the DPM Expression Language referenced in an expression as input.

The result produced by an *Operation* is also a specific artifact from the DPM Expression Language.

## 2.2.2 2.2 Language Artifacts

classDiagram class LanguageArtifact class Recordset class Scalar class ScalarSet class DataType Recordset --|> LanguageArtifact Scalar --|> LanguageArtifact ScalarSet --|> LanguageArtifact Recordset --> DataType : Defined on Scalar --> DataType : Defined on ScalarSet --> "1..n" Scalar style LanguageArtifact fill:#04f286 style Recordset fill:#04f286 style Scalar fill:#04f286 style ScalarSet fill:#04f286 style DataType fill:#ecdc0a

DPM Expression Language calculations can operate on and generate as results three different types of *Artifacts*.

#### 2.2.1 Scalars

*Scalars* are individual values of a certain *Data Type*.

#### 2.2.2 Scalar Sets

*Scalar Sets* are sets of *Scalar* values defined on the same *Data Type*. *Scalar Sets* are typically used with the in operator.

#### 2.2.3 Recordsets

classDiagram class Recordset class Records class Structure class Component class KeyComponent class FactComponent class AttributeComponent class DataType Recordset *-- Records Recordset --> Structure : Structured by Structure *-- Component Component --> DataType : Defined on KeyComponent --|> Component FactComponent --|> Component AttributeComponent --|> Component style Recordset fill:#04f286 style Records fill:#04f286 style Structure fill:#04f286 style Component fill:#04f286 style KeyComponent fill:#04f286 style FactComponent fill:#04f286 style AttributeComponent fill:#04f286 style DataType fill:#ecdc0a

*Recordsets* are collections of *Records* that share a same *Structure*. Technically, *Recordsets* are two-dimensional labelled data structures (tabular), which can be assimilated to Relational Tables or Data Frames. The columns (fields) of the *Recordset* are provided by the *Components* of its *Structure*. The rows of the *Recordset* are its composing *Records*.

The *Structure* of the *Recordset* is a collection of *Components*, which can have one of three roles: *Key*, *Fact* or *Attribute*. Each *Component* has a name, which must be unique within the *Recordset*. *Key* and *Attribute Components* are defined on a *Data Type*. *Fact Components* may be defined on more than one *Data Type*, when more than one *Variable* is selected. In any case, the *Data Type* is known for each selected *Variable*.

Each *Record* of the *Recordset* is individually identified by the combination of the values for its *Key Components*.

A *Recordset* having no *Key Components* behaves like a *Scalar*.

A *Recordset* may have no records.

#### 2.2.3.1 KEY COMPONENTS

classDiagram class KeyComponent class StandardKey class DpmKey class DpmXlStandardKey class DpmMlStandardKey class RowKey class ColumnKey class SheetKey class XIndexKey class YIndexKey class ZIndexKey StandardKey --|> KeyComponent DpmKey --|> KeyComponent DpmXlStandardKey --|> StandardKey DpmMlStandardKey --|> StandardKey RowKey --|> DpmXlStandardKey ColumnKey --|> DpmXlStandardKey SheetKey --|> DpmXlStandardKey XIndexKey --|> DpmMlStandardKey YIndexKey --|> DpmMlStandardKey ZIndexKey --|> DpmMlStandardKey style KeyComponent fill:#04f286 style StandardKey fill:#04f286 style DpmKey fill:#04f286 style DpmXlStandardKey fill:#04f286 style DpmMlStandardKey fill:#04f286 style RowKey fill:#04f286 style ColumnKey fill:#04f286 style SheetKey fill:#04f286 style XIndexKey fill:#04f286 style YIndexKey fill:#04f286 style ZIndexKey fill:#04f286

*Key Components* may be *Standard* or *DPM*. *Standard Key Components* are different for DPM-ML and DPM-XL, although their behaviour is the same.

*Standard Key Components* are common to all the *Recordsets*, independently on how the *Variables* are defined in the DPM. For each *Recordset*, there may be 0 or 1 occurrence of each subtype of *Standard Key Component*.

##### 2.2.3.1.1 Standard DPM-XL Key Components

- **Row Key**: Identifies the Row Ordinate from a Report Table where the selected *Variable* is located. Arises in *Variable Set* Selections, when more than one Row for one Report Table is selected. The name for the *Component* is "r". It is defined on the String *Data Type*.
- **Column Key**: Identifies the Column Ordinate from a Report Table where the selected *Variable* is located. Arises in *Variable Set* Selections, when more than one Column for one Report Table is selected. The name for the *Component* is "c". It is defined on the String *Data Type*.
- **Sheet Key**: Identifies the Sheet Ordinate from a Report Table where the selected *Variable* is located. Arises in *Variable Set* Selections, when more than one Sheet for one Report Table is selected. The name for the *Component* is "s". It is defined on the String *Data Type*.

##### 2.2.3.1.2 Standard DPM-ML Key Components

Regarding the DPM-ML, the *Standard Components* are three indexes, which do not represent any DPM object. They just serve to make sure that the right *Variables* are matched together when used with an *Operator*. The names for these *Components* are "x", "y" and "z". Note that these indexes do not necessarily correspond to rows, columns, and sheets. During the conversion from DPM-XL, the values of the indexes for each *Variable* shall be provided according to algorithms.

##### 2.2.3.1.3 DPM Key Components

*DPM Key Components* are specific to how data is defined in the DPM. Arise when Open *Variables* are selected, and a *Recordset* will have one *DPM Key Component* per each *Key Variable* associated to the selected *Variables*.

The name for the *DPM Key Components* is the Code of the Property associated to the DPM *Key Variable*.

#### 2.2.3.2 FACT COMPONENT

A *Recordset* has one mandatory, and only one, *Fact Component*. The name for the *Component* is "f". Its *Data Type* depends on the *Data Type* of the selected *Variables*.

If a DPM *Key Variable* is selected, the resulting *Recordset* will have the values for that *Variable* in the *Fact Component*, on top of having them in their corresponding *Key Component*. See example in section 2.2.4.4.

#### 2.2.3.3 ATTRIBUTE COMPONENTS

*Attribute Components* provide additional information to the *Fact* value.

DPM *Attribute Components* are specific to how data is defined in the DPM. Arise when a *Variable* with associated *Attribute Variable* is selected. A *Recordset* will have one DPM *Attribute Component* per each Property or Metric associated to any *Attribute Variable* associated to the selected *Variables*.

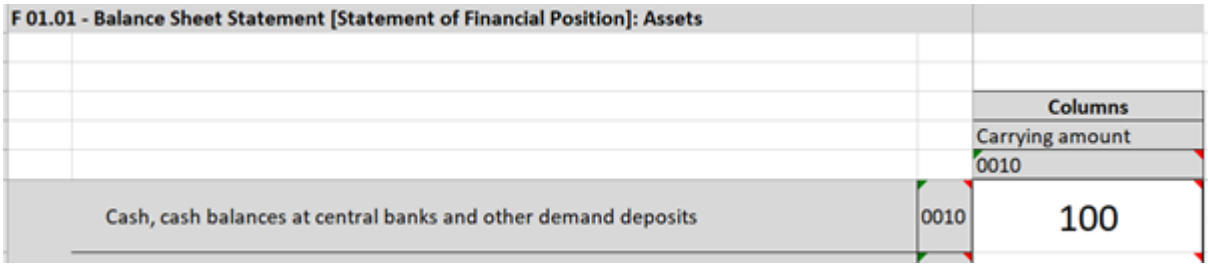
The name for the DPM *Attribute Components* is the Code of the Property associated to the DPM *Attribute Variable*.

### 2.2.4 Examples

*Recordsets* arise from Open *Variable* selections or from selection of multiple *Variables*.

#### 2.2.4.1 CLOSED VARIABLE SELECTION

For instance, taking as an example EBA's Table F 01.01:




```
{tF_01.01, r0010, c0010}
```

In the context of a single instance (i.e., disregarding the "ref_date" and the "subject"), selects a *Recordset* without *Key Components*, which works as a *Scalar* with value 100.

In the context of data with multiple dates and/or reference dates, and supposing that the data refer to the subject with code "id123" and to the date 31/12/2020, and that it has been reported with a 1 precision (attribute):

| subject | ref_date | f | p |
|---|---|---|---|
| id123 | 2020-12-31 | 100 | 1 |

#### 2.2.4.2 OPEN VARIABLE SELECTION

**F 40.01 - Scope of the group: “entity-by-entity”**

| | | | Columns | | |
|---|---|---|---|---|---|
| | | | Code | Type of code | Entity name |
| | | | 0011 | 0015 | 0031 |
| Rows | Investee | 999 | 123456 | LEI | Name1 |
| | | | 123456 | ISIN | Name2 |
| | | | 1111 | LEI | Name3 |
| | | | LIN <Key value> | TYC <Key value> | |

```
{F 40.01 c0031}
```

Selects an Open *Variable*. In this illustrative example, yields a *Recordset* with three *Records*. The *Recordset*, in the context of a single instance, has two DPM *Key Components*, because the selected *Variable* (the one rendered in Column 0031 of Report Table F 40.01) is associated in the DPM to a *Key* with two *Key Variables*. These *Key Variables* are associated to Properties with codes "LIN" (rendered in Column 0011) and "TYC" (rendered in Column 0015).

The following table represents the *Recordset*:

| LIN | TYC | f |
|---|---|---|
| 123456 | LEI | Name 1 |
| 123456 | ISIN | Name 2 |
| 1111 | LEI | Name 3 |

#### 2.2.4.3 VARIABLE SET SELECTION

**F 20.05.a - Geographical breakdown of off-balance sheet exposures by residence of the counterparty (a)**

**ES**

| | | | Columns |
|---|---|---|---|
| | | | Nominal amount |
| | | | 0010 |
| Rows | Loan commitments given | 0010 | 100 |
| | Financial guarantees given | 0020 | 200 |
| | Other commitments given | 0030 | 300 |

**PT**

| | | | Columns |
|---|---|---|---|
| | | | Nominal amount |
| | | | |
| | | | 0010 |
| Rows | Loan commitments given | 0010 | 400 |
| | Financial guarantees given | 0020 | 500 |
| | Other commitments given | 0030 | 600 |

```
{F 20.05 r0020-0030, c0010}
```

This case selects a set of *Variables* because the selection includes more than one Row. This implies that the *Row Key Component* applies to this *Recordset*. Besides, the selected *Variables* are associated to a *Key* with a single *Key Variable*, which is associated to a property with code "RCP".

| RCP | r | f |
|---|---|---|
| ES | 0020 | 200 |
| ES | 0030 | 300 |
| PT | 0020 | 500 |
| PT | 0030 | 600 |

In this case, the *Recordset* for the DPM-ML would be slightly different because of the change in the *Standard Key Component*. Instead of the row, we would have any of the three indexes. For instance, using the x index:

| RCP | x | f |
|---|---|---|
| ES | 1 | 200 |
| ES | 2 | 300 |
| PT | 1 | 500 |
| PT | 2 | 600 |

Note that the important thing for DPM-ML to be consistent is that all the *Records* with the same row share the same index.

#### 2.2.4.4 KEY VARIABLE SELECTION

**F 20.05.a - Geographical breakdown of off-balance sheet exposures by residence of the counterparty (a)**

**ES**

| | | | Columns |
|---|---|---|---|
| | | | Nominal amount |
| | | | 0010 |
| Rows | Loan commitments given | 0010 | 100 |
| | Financial guarantees given | 0020 | 200 |
| | Other commitments given | 0030 | 300 |

**PT**

| | | | Columns |
|---|---|---|---|
| | | | Nominal amount |
| | | | |
| | | | 0010 |
| Rows | Loan commitments given | 0010 | 400 |
| | Financial guarantees given | 0020 | 500 |
| | Other commitments given | 0030 | 600 |

```
{F 40.01 c0015}
```

Selects an Open *Key Variable*. The *Recordset*, in the context of a single instance, has two DPM *Key Components*, because the selected *Variable* (the one rendered in Column 0015 of Report Table F 40.01) is associated in the DPM to a *Key* with two *Key Variables* (in fact, it is part of the *Key*). These *Key Variables* are associated to Properties with codes "LIN" (rendered in Column 0011) and "TYC" (rendered in Column 0015).

The following table represents the *Recordset*:

| LIN | TYC | f |
|---|---|---|
| 123456 | LEI | LEI |
| 123456 | ISIN | ISIN |
| 1111 | LEI | LEI |

Note that the resulting *Recordset* has the values for the type of code two times, one as *Key Component* and another as *Fact Component*. This is necessary to allow doing calculations on the *Fact* while keeping all the *Records* in the *Recordset* uniquely identified.

## 2.2.3 2.3 Data Types

DPM *Operations* data types are defined with the principle: Create only a data type if they have different behaviour to other data types with at least one *Operator*.

The data types are:

- **String**: Sequence of alphanumeric characters of any length.
- **Number**: Is a rational number of any magnitude and precision, also used as approximation of a real number. Numbers can be treated as points or as intervals. Intervals are defined as a centre plus and minus a radius (c ± r). The centre is the input value provided. The radius is calculated based on the precision of the number as $10^{-p}/2$.
- **Integer**: Positive and negative integer numbers and zero. It Is a subtype of the type Number. As Number subtypes, Integers can be treated as intervals.
- **Time interval** denotes time intervals of any duration and expressed with a precision. According to ISO 8601 (ISO standard for the representation of dates and times), a time interval is the intervening time between two time points.
- **Date** is a subtype of the type Time which denotes time points expressed at any precision, which are time intervals starting and ending in the same time point (i.e., intervals of zero duration). A value of type Date includes all the parts needed to identify a time point at the desired precision, like the year, the month, the day, the hour, the minute and so on (for example, 2018-04-05 is the fifth of April 2018, at the precision of the day).
- **Time period** is a subtype of the type Time, and denotes non-overlapping time intervals having a regular duration (for example the years, the quarters of years, the months, the weeks and so on). A value of the type Time Period is composite and must include all the parts needed to identify a regular time period at the desired precision; in particular, the Time Period type includes the explicit indication of the kind of regular period considered (e.g., "day", "week", "month", "quarter" ...). For example, the value 2018M04, assuming that "M" stands for "month", denotes the month n.4 of the 2018 (April 2018). Moreover, 2018Q2, assuming that "Q" stands for "quarter", denotes the second quarter of 2018. In these examples, the letters M and Q are used to denote the kind of period through its duration.
- **Boolean**: denotes a logical binary state, meaning either `true` or `false`.
- **Category items**: A reference to the code of a Category Item.
- **Subcategories**: A reference to the code of a Subcategory.

#### 2.3.1 Correspondences with DPM Data Types

The DPM provides a wider list of data types. This is because greater granularity in the data typology may be useful for reporting or representation purposes (e.g., a monetary amount requires a currency attribute, while a pure number does not).

For each DPM data type there is one, and only one DPM-XL *Data type*.

| Code | Name | DPM-XL type |
|---|---|---|
| i | integer | Integer |
| r | decimal | Number |
| s | string (non empty) | String |
| b | boolean | Boolean |
| t | true | Boolean |
| dt | date time | Date |
| d | date | Date |
| e | enumeration | Category Items |
| m | monetary | Number |
| p | percentage | Number |
| u | URI | String |
| o | ordinals | Integer |
| es | string (including empty string) | String |

### 2.3.2 Casting

The casting between data types is possible. Casting can be done explicitly, if the cast *Operator* is used, or implicitly when it is allowed. The following table contains all the casting options.

| | String | Number | Integer | Time interval | Date | Time |
|---|---|---|---|---|---|---|
| **String** | | Explicit | Explicit | Explicit | Explicit | Explic |
| **Number** | Implicit | | Explicit | Not possible | Not possible | Not po |
| **Integer** | Implicit | Implicit | | Not possible | Not possible | Not po |
| **Time interval** | Implicit | Not possible | Not possible | | Explicit | Explic |
| **Date** | Implicit | Not possible | Not possible | Implicit | | Explic |
| **Time period** | Implicit | Not possible | Not possible | Implicit | Not possible | |
| **Duration** | Implicit | Not possible | Not possible | Not possible | Not possible | Not po |
| **Boolean** | Implicit | Not possible | Not possible | Not possible | Not possible | Not po |
| **Item** | Implicit | Not possible | Not possible | Not possible | Not possible | Not po |
| **Subcategory** | Implicit | Not possible | Not possible | Not possible | Not possible | Not po |

#### 2.3.3 Conventions to Write Scalars

To the effects of DPM-XL, when writing *Scalars*, the following conventions should be followed:

- **String**: Between double quotes ("this is a string") or single quotes ('this is also a string')
- **Number**: Sequence of numeric digits. Dot (".") shall be used to separate the decimal and the integer part.
- **Integer**: Sequence of numeric digits.
- **Time** Follows ISO 8601 (ISO standard for the representation of dates and times). Dates shall be written between hashes (#). A valid time representation is #2021-11-25/2021-12-25#
- **Date** #2021-11-25#.
- **Time period** #2021Q4#.
- **Duration** Following ISO8601: P[n]Y[n]M[n]DT[n]H[n]M[n]S. E.g.,: #P3Y6M4DT12H30M5S#
- **Boolean**: true or false.
- **DPM objects**: A string between brackets with two arguments, first the type and second the code. E.g., [item, eba_RT:x11], [subcategory, eba_CU:iso_currencies]

## 2.3 3 DPM-XL Data Selection

The data for the expression language is selected by referencing the DPM *Cells* and *Key Variables* of DPM Report Tables.

The references to the *Cells* and *Key Variables* are done by means of the table codes, ordinate codes and the codes of the Properties associated to the *Key Variables*.

The data selection is done by the interaction of the selection *Operator*, represented by the curly brackets symbol, and the general scope of an expression, provided by the with *Operator*. With this interaction, it is possible to individuate individual DPM *Variables* or sets of them. The actual data are referring to DPM *Variables*, thus making it possible to obtain the data required for the calculations.

The selection of data, together with the DPM definitions, make it possible to determine the *Structure* of the *Recordset* on compile-time.

# 2.4 4 Null in DPM Expression Language

## 2.4.1 4.1 Null and Data Types

All the *Data Types* are assumed to contain the conventional value null, which means "no value", or "absence of known value" or "missing value". Note that the null value, therefore, is the only value of multiple different types.

For the String *Data Type*, null is considered equivalent to the empty string.

## 2.4.2 4.2 Implicit and Explicit Null Values

For *Recordsets*, nulls may arise in a selection for two reasons:

- **Explicit null**: In the input data for the engine, there is a *Record* without a value.
- **Implicit null**: In the input data for the engine, there is no *Record* for one of the *Variables* in the selection. In this case, the engine will generate a *Record* for that *Variable* with a null value, or another specified default (see Selection *Operator* in part 2).

In any case, there are no differences in how implicit or explicit nulls are treated.

#### 4.2.1 Implicit Nulls Generation

Generation of implicit nulls is necessary because of the default values. Whenever a set of *Variables* is selected, all the selected *Variables* need to be created, even if they have not been reported (implicit null).

For **closed tables**, one *Record* shall be generated for each variable in the selection.

For **open tables**, for each combination of *Keys* that has at least one value reported (including explicit nulls), one *Record* for each non existing *Variable* shall be generated.

## 2.4.3 4.3 Null Treatment

In general, most of the *Operations* return null when any of their arguments is null.

- **Comparison Operators** (e.g., -, >): if a null is involved in the operation, then the result is null.
- **Arithmetic Operators** (e.g., + , -, *): if a null is involved in the operation, then the result is null.
- **String Operators**: null is considered an empty string.
- **Logical Operators** (and, or, xor, not): Three-value logic is adopted with the consideration that null means unknown. The concrete results for each *Operator* are specified in part 2, in the description of the *Operators*.
- **Conditional Operators**: null is considered equivalent to false.
- **Filtering**: null is considered equivalent to false (i.e., the *Records* with a null value are not selected in the filter).
- **Aggregations** (e.g., sum, avg, max): nulls are excluded from the calculations.
- **Intervals**: If the centre is null, the radius is also null.

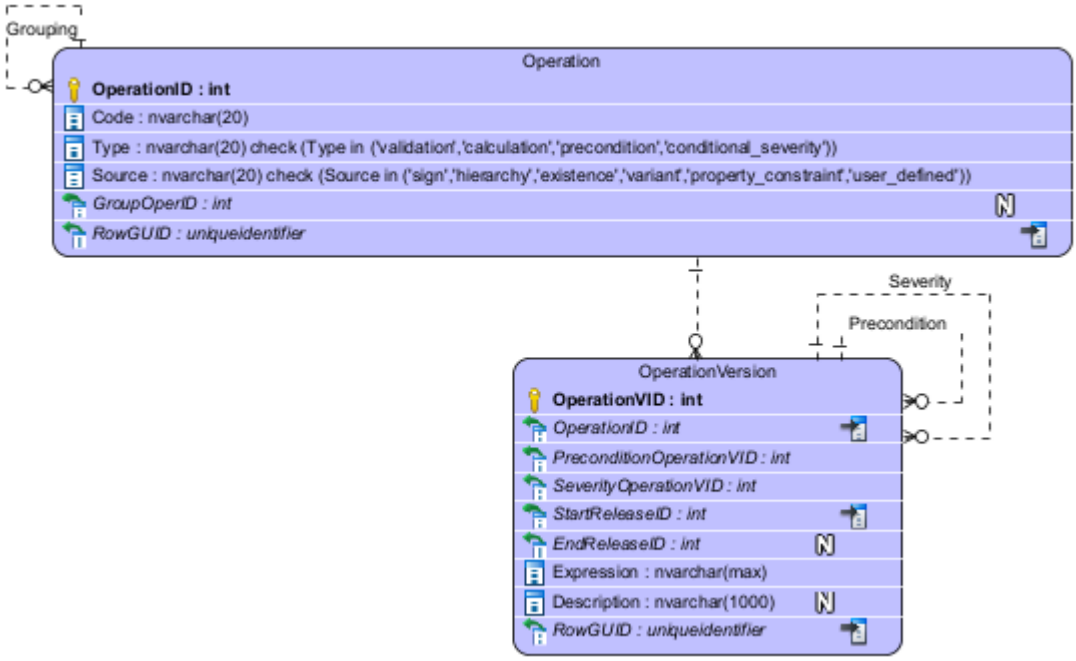
## 2.4.4 4.4 Overriding the Standard Null Treatment

The standard null treatment can be overridden by substituting the null values for other values. This can be done in two ways:

- **With the Selection Operator**: It is possible to set a default value in the selection for the null values.
- **With the nvl Operator**: serves to substitute nulls for any value.

# 2.5 5 DPM-ML Metamodel

## 2.5.1 5.1 Operation and OperationVersion




*Operations* have a code and can be grouped.

Groups of *Operations* refer to the cases where users define a parent validation and the system derives automatically children validations. There can only be two levels of *Operations* (i.e., a child validation cannot be the parent of another validation). The types of validations that generate children are Variant and Property constraint validations (see below).

*Operations* and may be of any of 4 types:

- **Validation**: *Operations* that serve to validate the consistency of data. They can provide as output a *Scalar* Boolean value or a *Recordset* with *Scalar* values.
- **Calculation**: *Operations* that serve to derive new data. They can provide as output any kind of data.
- **Precondition**: *Operations* that serve to determine whether one or more validations shall be applied. They can only provide as output a *Scalar* Boolean value.
- **Conditional_severity**: *Operations* that serve to determine the severity of an invalid result for a validation. It can yield a value 'error' or 'warning'

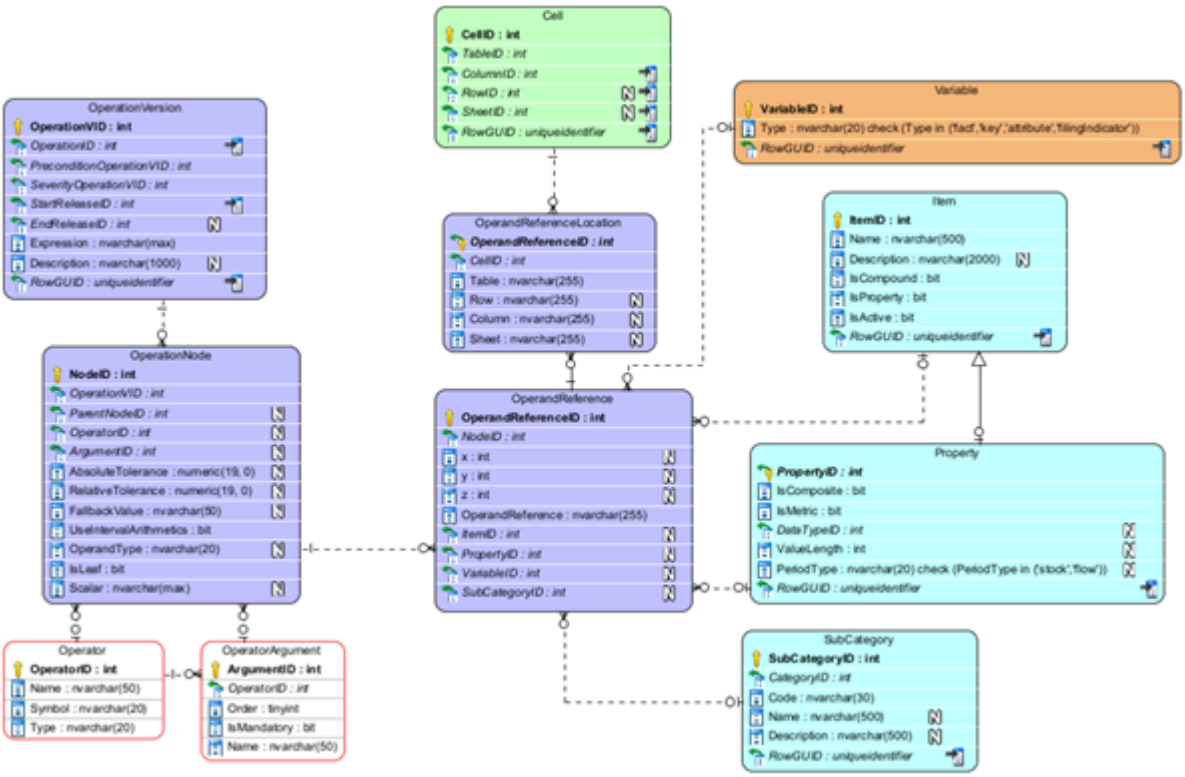
*Operations* are also classified based on their source for generation as:

- **User_defined**: *Operations* that have been defined by users.
- **Hierarchy**: Validations that have been automatically generated by the system based on the contents of subhierarchies.
- **Sign**: Validations that have been automatically generated by the system based on the contents of the Sign attribute for TableVersionCells.
- **Existence**: Validations that have been automatically generated by the system based on the IsNullable attribute for TableVersionCells.
- **Variant**: Validations that are defined at the level of TableGroups, including the individual validations generated from them.
- **Property_constraint**: Validations that are defined at the level of a property, and from which individual instances of validations based on *Variables* are generated.

*Operations* can have many versions. A new version of an *Operation* is required whenever any of the following attributes change:

- **Expression**: The DPM-XL expression of the version.
- **Description**: Natural language description of what the validation is aimed to check.
- **PreconditionOperation**: A link to an *Operation* version that acts as precondition.
- **SeverityOperation**: A link to an *Operation* that determines the conditional severity.

## 2.5.2 5.2 Operation Tree




### 5.2.1 Operation Nodes

The tree representation of an *OperationVersion* is provided in the *OperationNode* table.

Any *Operation* can be represented as a hierarchical tree, where the arguments of one *Operator* are represented.

For instance, the expression:

```
A = B + C
```

Can be represented as:

```
    =
   / \
  A   +
     / \
    B   C
```

Where "=" and "+" are *Operators*, and "A", "B" and "C" are *Operands*.

This kind of tree is represented in the *OperationNode* table.

### 5.2.2 *Operands*' Representation

All *Operands* are represented as *OperationNodes*.

*Operands* that are *Scalars* not representing a DPM Object (like an Item) are completely represented in the *OperationNode* table.

*Scalar Operands* that are referring to a DPM Object need to be further specified with the tables *OperandSource* and *OperandReference*, which provide the reference to the DPM Object.

*Operands* that are *Variables* and Sets of *Variables* need also to be further specified with the tables *OperandSource* and *OperandReference*. Concretely, for Sets of *Variables*, it is important to determine the index of each *Variable* in the context of the *Operand*, to be able to match the *Variables* to which the *Operator* applies.

#### 5.2.3 Representation Example

For instance, suppose a table (Tab1) with three rows (100, 200 and 300), two columns (100 and 200) and two sheets.

**Sheet A**

| | **Column 100** | **Column 200** |
|---|---|---|
| **Row 100** | 1 | 4 |
| **Row 200** | 2 | 5 |
| **Row 300** | 3 | 6 |

**Sheet B**

| | **Column 100** | **Column 200** |
|---|---|---|
| **Row 100** | 7 | 10 |
| **Row 200** | 8 | 11 |
| **Row 300** | 9 | 12 |

Note that each number in the cells represents the ID of the corresponding *Variable*.

The following expression:

```
{tTab1, r100, c*, s*} = 2 * ({tTab1, r200, c*, s*} + {tTab1, r300, c*, s*})
```

Would be represented as the following tree:

```
                =
              /   \
{tTab1, r100, c*, s*}   *
                      /  \
                     2    +
                        /  \
              {tTab1, r200, c*, s*} {tTab1, r300, c*, s*}
```

The representation in the DB would be as follows (only the relevant fields are shown, and for *Operators* and argument symbols are used):

**OperationNode Table**

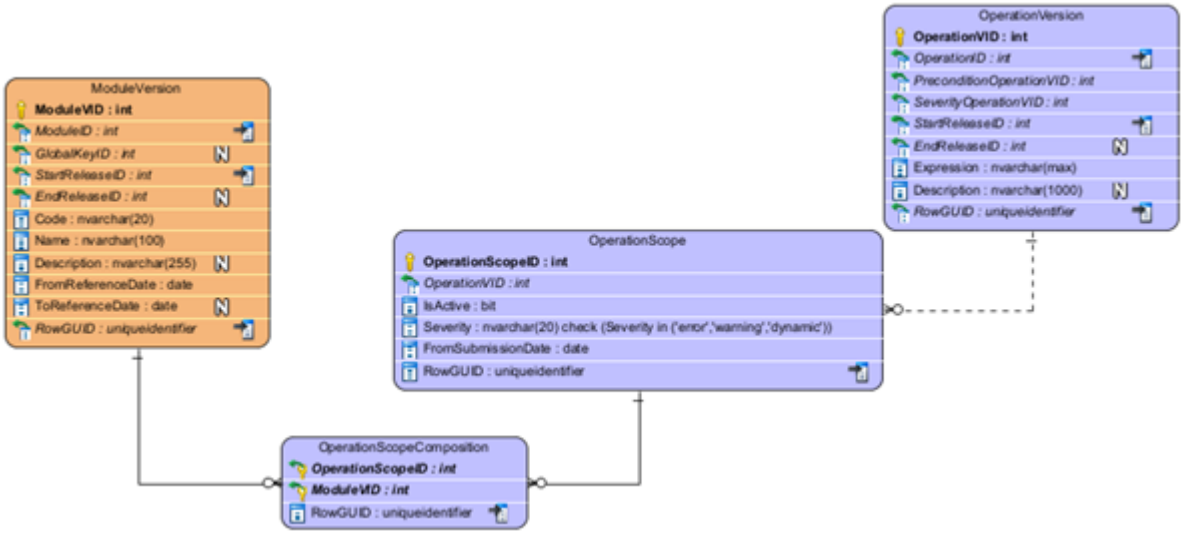
| NodeId | ParentNodeID | *Operator* | Argument | OperandSourceId | *Scalar* |
|---|---|---|---|---|---|
| 1 | | = | | | |
| 2 | 1 | | left | oprnd1 | |
| 3 | 1 | * | right | | |
| 4 | 3 | | left | | 2 |
| 5 | 3 | + | right | | |
| 6 | 5 | | left | oprnd2 | |
| 7 | 5 | | right | oprnd3 | |

**OperandReference Table**

| NodeID | x | y | z | VariableID |
|---|---|---|---|---|
| 2 | 1 | 1 | 1 | 1 |
| 2 | 2 | 1 | 1 | 4 |
| 2 | 1 | 1 | 2 | 7 |
| 2 | 2 | 1 | 2 | 10 |
| 6 | 1 | 1 | 1 | 2 |
| 6 | 2 | 1 | 1 | 5 |
| 6 | 1 | 1 | 2 | 8 |
| 6 | 2 | 1 | 2 | 11 |
| 7 | 1 | 1 | 1 | 3 |
| 7 | 2 | 1 | 1 | 6 |
| 7 | 1 | 1 | 2 | 9 |
| 7 | 2 | 1 | 2 | 12 |

Note that the *Operations* shall be applied to the *Variables* having the same index. For instance, *Variable* 1 = 2 * (*Variable* 2 + *Variable* 3), because all of them have the index y=1 and z=1.

## 2.5.3 5.3 Operation Scope

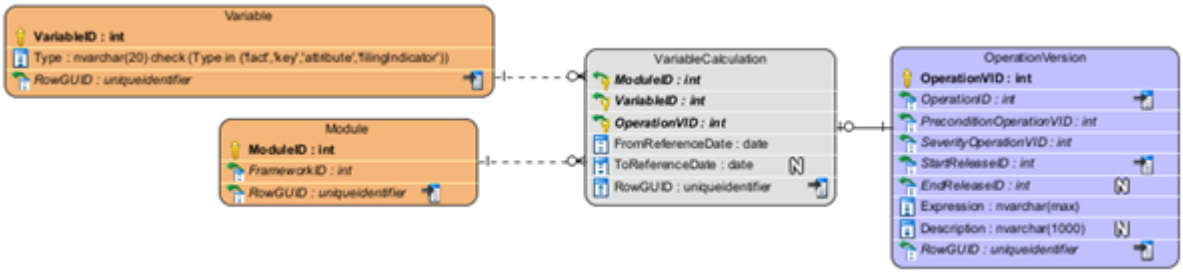



Each *Operation* version can be applied in different scopes. An *Operation* scope refers to the individual module versions or sets of module versions (for cross-module validations) to which the *Operation* logically applies.

For each *Operation* scope, there may be different values for the attributes:

- **IsActive**: Determines whether the *Operation* shall be run for a certain scope.
- **Severity**: Determines the severity of the error, if the validation is not passed.
- **FromSubmissionDate**: Sets a date from which the validation applies to subsequent submissions.

## 2.5.4 5.4 Variable Calculation




For calculations (i.e., *Operations* that serve to calculate values for *Variables*), it is necessary to link the *Operation* to the actual *Variable* that is generated. This is done with the *VariableCalculation* table, which links to the *Variable* and the Module. The link to the Module is necessary due to the fact that a *Variable* can be used in several Modules, being calculated in some cases and not calculated in other cases.

# 3. Operators

## 3.1 1 Introduction

This documentation provides the individual description of the semantics for all the *Operators* in the DPM Operations (DPM-XL and DPM-ML).

The document is structured from more generic to more concrete. It starts with general behaviours that apply to several *Operators* of different types. Then it follows with the common behaviour of the *Operators* of the same type, finalizing with the specific behaviour of the single *Operators*.

# 3.2 2 Common General Behaviours for the Operators

## 3.2.1 2.1 Unary Operators

This general behaviour is applied to *Operators* taking as an argument one single *Operand*.

Unless explicitly specified differently, the behaviour for the unary *Operators* is as follows:

#### 2.1.1 Scalars

The *Operator* is applied on a *Scalar* value and returns a *Scalar* value.

#### 2.1.2 Recordsets

The *Operator* is applied on a *Recordset* and returns a *Recordset*.

The *Operator* is applied to the values of all the facts of the *Recordset*.

##### 2.1.2.1 CONSTRAINTS

1. The application of the *Operator* is only allowed if all the facts of the *Operand Recordset* are compatible with the *Operator*.

##### 2.1.2.2 STRUCTURE

The structure of the resulting *Recordset* contains the *Key* and *Fact Components* of the *Operand Recordset*, the *Attribute Components* are not propagated.

##### 2.1.2.3 RECORDS

The *Operators* produce one output *Record* per each input *Record*, which have:

- For *Key Components*, the values unchanged.
- For the *Fact* values, the *Operator* is applied to the input value and returns the corresponding value.

## 3.2.2 2.2 Binary Operators

This general behaviour is applied whenever an *Operator* takes as input two *Operands*, which includes the following *Operators*:

- Arithmetic Binary Operators
- Comparison Operators
- Logical Binary Operators

Unless explicitly specified differently, the behaviour for the binary *Operators* is as follows.

#### 2.2.1 Scalars

If the two *Operands* of a binary *Operator* are *Scalars*, the result shall be the *Scalar* resulting of applying the *Operator* to the *Operands*.

#### 2.2.2 Recordset and Scalar

A binary *Operator* applied to a *Recordset Operand* and a *Scalar*, will result in a *Recordset* with the same *Key* and *Fact Components* as the input *Recordset Operand* (*Attribute Components* are not propagated). The *Operator* shall be applied to every *Record* of the input *Recordset* and the *Scalar*.

##### 2.2.2.1 EXAMPLES

```
0.25*{S.26.01, r0600, cNNN}
```

Supposing that the selection yields the following *Recordset*:

| c | f |
|---|---|
| 0060 | 100 |
| 0080 | 200 |

The operation results in:

| c | f |
|---|---|
| 0060 | 25 |
| 0080 | 50 |

### 2.2.3 Recordsets

#### 2.2.3.1 CONSTRAINTS

Binary *Operators* can only be applied to two *Recordsets Operands* if they have:

1. exactly the same *Key Components*; or
2. the *Key Components* of one *Recordset* (Reference *Recordset*) are a superset of the *Key Components* of the other *Recordset*.
3. *Standard Keys* contain the same values. E.g., summing `{tT1, r010-020}` and `{tT1, r030-040}` is not possible, because there are no matches in the operands.

#### 2.2.3.2 STRUCTURE

The *Operator* yields a *Recordset* with the common *Key Components* in case 1, or the *Key Components* of the Reference *Recordset* in case 2. The resulting *Recordset* does not contain any *Attribute Component* from the *Operand Recordsets*.

#### 2.2.3.3 RECORDS

The *Operator* applies to the pairs of values resulting from performing an inner join of the input *Recordsets* on the common *Key Components*.

#### 2.2.3.4 EXAMPLES

##### 2.2.3.4.1 Same Identifiers

```
{C 28.00 c040} + {C 28.00 c190}
```

Supposing that the selections yield:

{C 28.00 c040}

| INC | f |
|---|---|
| 123 | 1000 |
| 456 | 2000 |
| 789 | 3000 |

{C 28.00 c190}

| INC | f |
|---|---|
| 123 | -100 |
| 456 | -200 |
| 789 | -300 |

The result would be:

| INC | f |
|---|---|
| 123 | 900 |
| 456 | 1800 |
| 789 | 2700 |

##### 2.2.3.4.2 Subset of Identifiers

```
{F 40.01 c0110} >= {F 40.02c0060}
```

Supposing that the selections yield:

{F 40.01 c0110}

| LIN | TYC | f |
|---|---|---|
| 123 | x1 | 1 |
| 456 | x1 | 0.8 |
| 789 | x1 | 0.4 |

{F 40.02 c0060}

| LIN | TYC | STC | LHC | LHO | f |
|---|---|---|---|---|---|
| 123 | x1 | 111 | ABC | x1 | 0.3 |
| 123 | x1 | 111 | DEF | x1 | 0.7 |
| 456 | x1 | 222 | ABC | x1 | 0.85 |

The result would be:

| LIN | TYC | STC | LHC | LHO | f |
|---|---|---|---|---|---|
| 123 | x1 | 111 | ABC | x1 | true |
| 123 | x1 | 111 | DEF | x1 | true |
| 456 | x1 | 222 | ABC | x1 | false |

Note that:

- An inner join on the common *Key Components* is performed. This means that there is no comparison for the *Record* in F 40.01 with LIN = 789, because there is no match in F 40.02.
- The result of the operation has the *Structure* of the Reference *Recordset*, which is the one that has a superset of *Key Components*.
- For each match in the join the *Operator* `>=` is applied to the pairs of values for the facts.

### 2.2.4 Intervals Treatment

When one of two numeric *Operands* is an interval, and the other is a point, the point is transformed into an interval with centre value equal to the point value, and radius 0.

# 3.3 3 Selection Operators

## 3.3.1 3.1 Selection Operator

#### 3.1.1 Syntax

{

[**t**table | **v**variable | **o**operation]1

[**r**row [, **r**row] | *rrow_init–row_end* | *r* ]*

[**c**col [, **c**col] | *ccol_init–col_end* | *c* ]*

[**s**sheet [, **s**sheet] | *ssheet_init–sheet_end* | *s* ]*

*interval_arithmetics*, *fallback_value*

}

#### 3.1.2 Input Parameters

- **table**: The code of a DPM Report table.
- **variable**: the code of a DPM *Variable*.
- **operation**: The code of a DPM *Operation*.
- **row, row_init, row_end**: A Reference to the code of a row of a DPM table.
- **col, col_init, col_end**: A Reference to the code of a col of a DPM table.
- **sheet, sheet_init, sheet_end**: A Reference to the code of a sheet of a DPM table.
- **interval_arithmetics**: A Boolean value specifying whether interval arithmetics should apply. The default value, in case the parameter is omitted, is false.
- **default_value**: A *Scalar* value specifying the value to be used in case of nulls.

#### 3.1.3 Output

rset<*>

#### 3.1.4 Semantics

Serves to select data as defined in the DPM model.

#### 3.1.5 Additional Constraints

1. Existence of the parameters. The referred table, *Variable* or *Operation* need to be defined in the DPM.
2. *Operations* need to be defined in the same script.
3. Rows, columns and sheets need to exist in the table or *Operation* on which they are defined.
4. When the selection is done at *Variable* level, row, column and sheet parameters cannot exist.

#### 3.1.6 Behaviour

Returns a *Recordset* with the following structure:

*Key Components*:

- DPM-XL
- Row: If the selected table has ordinate rows and more than one row is present in the selection. The name of the *Component* is "r".
- Column: If the selected table has ordinate columns and more than one column is present in the selection. The name of the *Component* is "c".
- Sheet: If the selected table has ordinate sheets and more than one sheet is present in the selection. The name of the *Component* is "s".
- DPM-ML
- X: If there is Row for DPM-XL.
- Y: If there is Column for DPM-XL.
- Z: If there is Sheet for DPM-XL.
- A DPM *Key Component* for each *Key Variable* belonging to the *Key* associated to the selected *Variable*. The name of the *Component* is the *Code* of the Property associated to the *Key Variable*. The *Data Type* of the *Component* shall be the correspondent *Data Type* of the Metric or Property associated to the respective *Key Variable*.

*Fact Component*: With the *Data Type* corresponding to the selected *Variable*.

*Attribute Components*: One *Component* for each *Attribute Variable* associated to the selected *Variable*. The name of the *Component* is the *Code* of the Property associated to the *Attribute Variable*. The *Data Type* of the *Component* shall be the correspondent *Data Type* of the Metric or Property associated to the respective *Attribute Variable*.

The *Records* for the *Recordset* shall be obtained from the input data according to the selection.

For ranges (e.g., r0010-0100), the selection is done based on the alphabetical order of the DPM *Header* codes. In the example, it selects all the *Headers* with a code between 0010 and 0100.

#### 3.1.7 DPM-ML Metamodel Representation

There is no metamodel representation for the selection *Operator*. It is used to select, in last instance, *Variables*. So the result of the selection are processed to get the relevant *Variables* for DPM-ML, as well as the *DefaultValue* and *UseIntervalArithmetics* attributes.

## 3.3.2 3.2 With

#### 3.2.1 Syntax

**with** partial_selection: *expression*

#### 3.2.2 Input Parameters

- **partial_selection**: A selection expression (see Selection *Operator*)
- **expression**: A expression including selection *Operators*

#### 3.2.3 Output

Does not generate output; modifies the selections of the expression after the colon.

#### 3.2.4 Semantics

Serves to simplify expressions by adding a single context that may apply to all the *Operands* in the expression.

#### 3.2.5 Additional Constraints

None

#### 3.2.6 Behaviour

The selection parameters in the partial selection applies to all the selections in the expression, unless they are overridden by a more specific parameter.

Therefore, in a selection inside the expression, whenever one of the parameters of the selection expression is not present, but that parameter is present on the partial selection, the parameter of the partial selection applies.

#### 3.2.7 Examples

```
With {F 01.01 c0010, default:0, interval:false}:
{r0010} = {r0020} + {r0030} + {r0040}
```

The partial selection applies to all the selections in the expression

```
With {F 01.01 c0010, default:0, interval:false}:
{r0010} + {r0040} = {F04.01 r0010, c0010}
```

The partial selection applies to the two selections that do not refer to a table. Regarding the other selection, the default and intervals apply

```
With {F 01.01 c0010, default:0, interval:false}:
{F 01.01 r0010} + {F 01.01 r0040} = {F04.01 r0010, c0010 default:null}
```

The partial selection applies to the two selections that refer to the same table referred to in the partial selection. Regarding the other selection, interval argument in the partial selection applies, but the default is overridden.

```
With {c0010, default:0, interval:false}:
{F 01.01 r0010} + {F 01.01 r0040} = {F04.01 r0010}
```

The partial selection applies to all the selections in the expression.

#### 3.2.8 Metamodel Representation

See selection *Operator*.

# 3.4 4 Numeric Operators

Numeric *Operators* describe operations involving *Operands* with data type Number, Integer or Number Interval.

## 3.4.1 4.1 Numeric Operators' General Behaviour

#### 4.1.1 Input Parameters

*Operands* can be of *Recordset* or *Scalar* type. Must be defined as *Number interval*, *Number*, or *Integer*.

#### 4.1.2 Result Type

*Recordset* or *Scalar* with type *Number interval*, *Number*, or *Integer*.

#### 4.1.3 Behaviour

If any *Operand* is null, then the result is also null.

Numeric *Operators* can be applied to any numeric type (*Number interval*, *Number*, or *Integer*) and combination of them, in which case casting to the highest type shall apply.

If the type of *Operands* is *Integer* then the result has type *Integer*. If any of the *Operands* is of type *Number*, and there are no *Number intervals*, then the result has type *Number*. If any *Operand* is of *Number Interval* type, the result has type Number Interval.

## 3.4.2 4.2 Unary Arithmetic Operators

#### 4.2.1 Plus (+)

##### 4.2.1.1 SYNTAX

```
+ op
```

##### 4.2.1.2 INPUT PARAMETERS

Op: rset | scal <num + interval>

##### 4.2.1.3 OUTPUT

rset | scal <num>

##### 4.2.1.4 SEMANTICS

Returns the *Operand* unchanged.

##### 4.2.1.5 ADDITIONAL CONSTRAINTS

None

##### 4.2.1.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.

##### 4.2.1.7 EXAMPLES

- `+ 1` results in `1`
- `+ (-3)` results in `-3`

### 4.2.2 Minus (-)

#### 4.2.2.1 SYNTAX

- op

#### 4.2.2.2 INPUT PARAMETERS

Op: rset | scal <num + interval>

#### 4.2.2.3 OUTPUT

rset | scal <num>

#### 4.2.2.4 SEMANTICS

Inverts the sign of the *Operand*.

For intervals, inverts the sign of the centre, leaving the radius unchanged.

#### 4.2.2.5 ADDITIONAL CONSTRAINTS

None

#### 4.2.2.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.

#### 4.2.2.7 EXAMPLES

- `- 1` results in `-1`
- `- (-3)` results in `3`

### 4.2.3 Absolute Value (abs)

#### 4.2.3.1 SYNTAX

**abs(** op **)**

#### 4.2.3.2 INPUT PARAMETERS

Op: rset | scal <num + interval>

#### 4.2.3.3 OUTPUT

rset | scal <num>

#### 4.2.3.4 SEMANTICS

Calculates the absolute value of a Number.

For intervals, return the absolute value of the centre, leaving the radius unchanged

#### 4.2.3.5 ADDITIONAL CONSTRAINTS

None

#### 4.2.3.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.

#### 4.2.3.7 EXAMPLES

- `Abs(-1)` results in `1`
- `Abs(3)` results in `3`

### 4.2.4 Exponential (exp)

#### 4.2.4.1 SYNTAX

**exp(** op **)**

#### 4.2.4.2 INPUT PARAMETERS

Op: rset | scal <num>

#### 4.2.4.3 OUTPUT

rset | scal

#### 4.2.4.4 SEMANTICS

Returns `e` (base of the natural logarithm) raised to the op power.

#### 4.2.4.5 ADDITIONAL CONSTRAINTS

None

#### 4.2.4.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.

#### 4.2.4.7 EXAMPLES

- `exp(2)` results in `7.38905`
- `exp(1)` results in `2.71828` (the `e` number)

### 4.2.5 Natural Logarithm (ln)

#### 4.2.5.1 SYNTAX

**ln(** op **)**

#### 4.2.5.2 INPUT PARAMETERS

Op: rset | scal <num>

#### 4.2.5.3 OUTPUT

rset | scal <num>

#### 4.2.5.4 SEMANTICS

Calculates the natural logarithm of a Number.

#### 4.2.5.5 ADDITIONAL CONSTRAINTS

The numeric values must be greater than 0.

#### 4.2.5.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.
- If any value is smaller than or equal to 0, generates a run-time error.

#### 4.2.5.7 EXAMPLES

- `ln(1)` results in `0`
- `ln(148)` results in `4.997`

### 4.2.6 Square Root (sqrt)

#### 4.2.6.1 SYNTAX

**sqrt(** op **)**

#### 4.2.6.2 INPUT PARAMETERS

Op: rset | scal <num>

#### 4.2.6.3 OUTPUT

rset | scal <num>

#### 4.2.6.4 SEMANTICS

Calculates the square root of a Number.

#### 4.2.6.5 ADDITIONAL CONSTRAINTS

The numeric values must be greater than or equal to 0.

#### 4.2.6.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.
- If any value is smaller than 0, generates a run-time error.

#### 4.2.6.7 EXAMPLES

- `sqrt(4)` results in `2`
- `sqrt(25)` results in `5`

### 4.2.7 Logarithm (log)

#### 4.2.7.1 SYNTAX

**log(** op, base **)**

#### 4.2.7.2 INPUT PARAMETERS

op: rset | scal <num>

base: scal <num>

#### 4.2.7.3 OUTPUT

rset | scal <num>

#### 4.2.7.4 SEMANTICS

Calculates the logarithm of base op.

#### 4.2.7.5 ADDITIONAL CONSTRAINTS

op numeric values must be greater than 1. base numeric values must be greater than 0.

#### 4.2.7.6 BEHAVIOUR

- Unary *Operators'* standard behaviour.
- Numeric *Operators'* standard behaviour.
- If the base is 1 or smaller, generates a run-time error.
- If the number is 0 or smaller, generates a run-time error.

#### 4.2.7.7 EXAMPLES

- `log(512, 2)` results in `9`

- `log(100, 10)` results in `2`

## 3.4.3 4.3 Binary Arithmetic Operators

This *Operators* group follows the General behavior for binary *Operators*.

### 4.3.1 Addition (+)

#### 4.3.1.1 SYNTAX

left + right

#### 4.3.1.2 INPUT PARAMETERS

left: rset | scal <num + interval>

right: rset | scal <num + interval>

#### 4.3.1.3 OUTPUT

rset | scal <num + interval>

#### 4.3.1.4 SEMANTICS

Returns the sum of two Numbers.

For intervals:

- Centre is calculated as `centre(left) + centre(right)`
- Radius is calculated `radius(left) + radius(right)`

#### 4.3.1.5 ADDITIONAL CONSTRAINTS

None.

#### 4.3.1.6 BEHAVIOUR

- Binary *Operators'* standard behaviour
- Numeric *Operators'* standard behaviour.

#### 4.3.1.7 EXAMPLES

- `3 + 2` results in `5`
- `-7 + 3` results in `-4`

### 4.3.2 Subtraction (-)

#### 4.3.2.1 SYNTAX

left - right

#### 4.3.2.2 INPUT PARAMETERS

left: rset | scal <num + interval>

right: rset | scal <num + interval>

#### 4.3.2.3 OUTPUT

rset | scal <num + interval>

#### 4.3.2.4 SEMANTICS

Returns the difference of two Numbers.

For intervals:

- Centre is calculated as `centre(left) - centre(right)`
- Radius is calculated `radius(left) + radius(right)`

##### 4.3.2.5 ADDITIONAL CONSTRAINTS

None.

##### 4.3.2.6 BEHAVIOUR

- Binary *Operators'* standard behaviour
- Numeric *Operators'* standard behaviour.

##### 4.3.2.7 EXAMPLES

- `3 - 2` results in `1`
- `-7 - 3` results in `-10`

### 4.3.3 Multiplication (*)

##### 4.3.3.1 SYNTAX

left * right

##### 4.3.3.2 INPUT PARAMETERS

left: rset | scal <num + interval>

right: rset | scal <num + interval>

##### 4.3.3.3 OUTPUT

rset | scal <num + interval>

##### 4.3.3.4 SEMANTICS

Returns the product of two Numbers.

For intervals:

- Centre is calculated as `centre(left) * centre(right)`
- Radius is calculated as `(centre(left) * radius(right)) + abs(radius(left) * centre(right)) + (radius(left) * radius(right))`

##### 4.3.3.5 ADDITIONAL CONSTRAINTS

None.

##### 4.3.3.6 BEHAVIOUR

- Binary *Operators'* standard behaviour
- Numeric *Operators'* standard behaviour.

##### 4.3.3.7 EXAMPLES

- `4 * 6` results in `24`
- `-9 * 2` results in `-18`

### 4.3.4 Division (/)

##### 4.3.4.1 SYNTAX

num / den

#### 4.3.4.2 INPUT PARAMETERS

num: rset | scal <num>

den: rset | scal <num>

#### 4.3.4.3 OUTPUT

rset | scal <num>

#### 4.3.4.4 SEMANTICS

Divides two Numbers.

For intervals: - Centre is calculated as `centre(left) / centre(right)` - Radius is calculated as:

```
max(
      abs(centre(left) + radius(left)) / (centre(right) + radius(right)),
      abs(centre(left) + radius(left)) / (centre(right) - radius(right)),
      abs(centre(left) - radius(left)) / (centre(right) + radius(right)),
      abs(centre(left) - radius(left)) / (centre(right) - radius(right))
)
```

#### 4.3.4.5 ADDITIONAL CONSTRAINTS

The denominator must be different to zero.

#### 4.3.4.6 BEHAVIOUR

- Binary *Operators'* standard behaviour
- Numeric *Operators'* standard behaviour.
- If the denominator is 0, generates a run-time error.

#### 4.3.4.7 EXAMPLES

- `24 / 6` results in `4`
- `-18 / 2` results in `-9`

### 4.3.5 Maximum (max)

#### 4.3.5.1 SYNTAX

**max(** op1, op2 {, op} *)*

#### 4.3.5.2 INPUT PARAMETERS

op1: rset | scal <num + interval>

op2: rset | scal <num + interval>

#### 4.3.5.3 OUTPUT

rset | scal <num + interval>

#### 4.3.5.4 SEMANTICS

Calculates the maximum value from a set of *Operands*.

For intervals: - Centre is calculated as the maximum centre value from all the intervals. - Radius is the radius of the interval with maximum centre.

#### 4.3.5.5 ADDITIONAL CONSTRAINTS

None.

#### 4.3.5.6 BEHAVIOUR

- For each pair of *Operands*, the standard behaviour for binary *Operators* applies.

##### 4.3.5.7 EXAMPLES

- `Max(1, 3, -5)` results in `3`
- `Max(1, 3, null)` results in `null`

### 4.3.6 Minimum (min)

##### 4.3.6.1 SYNTAX

**min(** op1, op2 {, op} **)***

##### 4.3.6.2 INPUT PARAMETERS

op1: rset | scal <num + interval>

op2: rset | scal <num + interval>

##### 4.3.6.3 OUTPUT

rset | scal <num + interval>

##### 4.3.6.4 SEMANTICS

Calculates the minimum value from a set of *Operands*.

For intervals:

- Centre is calculated as the minimum centre value from all the intervals.
- Radius is the radius of the interval with minimum centre.

##### 4.3.6.5 ADDITIONAL CONSTRAINTS

None.

##### 4.3.6.6 BEHAVIOUR

- For each pair of *Operands*, the standard behaviour for binary *Operators* applies.

##### 4.3.6.7 EXAMPLES

- `Min(1, 3, -5)` results in `-5`
- `Min(1, 3, null)` results in `null`

### 4.3.7 Power

##### 4.3.7.1 SYNTAX

**power(** base, exponent **)**

##### 4.3.7.2 INPUT PARAMETERS

num: rset | scal <num>

den: rset | scal <num>

##### 4.3.7.3 OUTPUT

rset | scal <num>

##### 4.3.7.4 SEMANTICS

Raises the power to the exponent.

##### 4.3.7.5 ADDITIONAL CONSTRAINTS

None.

#### 4.3.7.6 BEHAVIOUR

- Binary *Operators'* standard behaviour
- Numeric *Operators'* standard behaviour.

#### 4.3.7.7 EXAMPLES

- `power(5,2)` results in `25`
- `power(5,-1)` results in `0.2`
- `power(-5, 3)` results in `-125`

# 3.5 5 Comparison Operators

## 3.5.1 5.1 Comparison Operators' General Behaviour

Comparison *Operators* describe operations that compare the values of *Operands*.

This *Operator* group is based upon the General behavior for binary *Operators*.

#### 5.1.1 Input Parameters

*Operands* that can be of *Recordset* or *Scalar* type. Must have the same *Data Type*. For all *Operators* that accept numeric *Operands*, intervals are allowed.

#### 5.1.2 Result Type

*Recordset* or *Scalar* with type Boolean.

#### 5.1.3 Constraints

The *Operands* for the comparison operations must be of the same type (considering implicit casting).

#### 5.1.4 Behaviour

If any *Operand* is null, then the result is also null.

For comparison *Operators* implying an order (`>` , `>=` , `<` , `<=` ), the following rules apply:

- Boolean values: `true` is considered greater than `false` .
- Strings: Alphabetic order is followed.

## 3.5.2 5.2 Equal (=)

#### 5.2.1.1 Syntax

left = right

#### 5.2.1.2 Input Parameters

left: rset | scal <*>

right: rset | scal <*>

#### 5.2.1.3 Output

rset | scal <boo>

#### 5.2.1.4 Semantics

Returns `true` if left is equal to right and `false` otherwise.

For intervals: `abs(centre(left) – centre(right)) <= radius(left) + radius(right)` .

#### 5.2.1.5 Additional Constraints

None.

#### 5.2.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.2.1.7 Examples

- `1 = 2` results in `false`
- `3 = null` results in `null`
- `"4" = "4"` results in `true`

## 3.5.3 5.3 Not Equal (!=)

#### 5.3.1.1 Syntax

left != right

#### 5.3.1.2 Input Parameters

left: rset | scal <*>

right: rset | scal <*>

#### 5.3.1.3 Output

rset | scal <boo>

#### 5.3.1.4 Semantics

Returns `false` if left is equal to right and `true` otherwise.

For intervals: `abs(centre(left) - centre(right)) > radius(left) + radius(right)`.

#### 5.3.1.5 Additional Constraints

None.

#### 5.3.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.3.1.7 Examples

- `1 != 2` results in `true`
- `3 != null` results in `null`
- `"4" != "4"` results in `false`

## 3.5.4 5.4 Greater than (>)

#### 5.4.1.1 Syntax

left > right

#### 5.4.1.2 Input Parameters

left: rset | scal <*>

right: rset | scal <*>

#### 5.4.1.3 Output

rset | scal <boo>

#### 5.4.1.4 Semantics

Returns `true` if left is greater than right and `false` otherwise.

For intervals: `centre(left) > centre(right) - (radius(left) + radius(right))`.

#### 5.4.1.5 Additional Constraints

None.

#### 5.4.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.4.1.7 Examples

- `12 > 2` results in `true`
- `3 > null` results in `null`
- `true > false` results in `true`
- `"tez" > "test"` results in `true`

## 3.5.5 5.5 Greater Equal than (>=)

#### 5.5.1.1 Syntax

left >= right

#### 5.5.1.2 Input Parameters

left: rset | scal <*>

right: rset | scal <*>

#### 5.5.1.3 Output

rset | scal <boo>

#### 5.5.1.4 Semantics

Returns `true` if left is greater than or equal to right and `false` otherwise.

For intervals: `centre(left) >= centre(right) - (radius(left) + radius(right))`.

#### 5.5.1.5 Additional Constraints

None.

#### 5.5.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.5.1.7 Examples

- `1 >= 2` results in `false`
- `3 >= null` results in `null`
- `"tez" >= "test"` results in `true`

## 3.5.6 5.6 Less than (<)

#### 5.6.1.1 Syntax

left < right

#### 5.6.1.2 Input Parameters

left: rset | scal <*>

right: rset | scal <*>

#### 5.6.1.3 Output

rset | scal <boo>

#### 5.6.1.4 Semantics

Returns `true` if left is smaller than right and `false` otherwise.

For intervals: `centre(left) - centre(right) < radius(left) + radius(right)`.

#### 5.6.1.5 Additional Constraints

None.

#### 5.6.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.6.1.7 Examples

- `1 < 2` results in `true`
- `3 < null` results in `null`
- `"tea" < "test"` results in `true`

## 3.5.7 5.7 Less Equal than (<=)

#### 5.7.1.1 Syntax

left <= right

#### 5.7.1.2 Input Parameters

left: rset | scal <*>

right: rset | scal <*>

#### 5.7.1.3 Output

rset | scal <boo>

#### 5.7.1.4 Semantics

Returns `true` if left is smaller than or equal to right and `false` otherwise.

For intervals: `centre(left) - centre(right) <= radius(left) + radius(right)`.

#### 5.7.1.5 Additional Constraints

None.

#### 5.7.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.7.1.7 Examples

- `3 <= 2` results in `false`
- `3 <= null` results in `null`
- `"tea" <= "test"` results in `true`

## 3.5.8 5.8 Element of (in)

#### 5.8.1.1 Syntax

op **in** set

#### 5.8.1.2 Input Parameters

op: rset | scal <num>

set: scalar_set <*> | subcategory

#### 5.8.1.3 Output

rset | scal <boo>

#### 5.8.1.4 Semantics

Returns `true` if op belongs to the set and `false` otherwise.

#### 5.8.1.5 Additional Constraints

op must be of the same *Data Type* as the values in the set (considering implicit casting)

#### 5.8.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.8.1.7 Examples

- `5 in {1,3, 5}` results in `true`
- `"abc" in {"def", "ghi"}` results in `false`

## 3.5.9 5.9 Match Characters (match)

#### 5.9.1.1 Syntax

**match(** op, pattern **)**

#### 5.9.1.2 Input Parameters

op: rset | scal <str>

pattern: scal <str>

#### 5.9.1.3 Output

rset | scal <boo>

#### 5.9.1.4 Semantics

Returns true if op matches the pattern, false otherwise.

#### 5.9.1.5 Additional Constraints

pattern is a regex expression following the Python definition.

#### 5.9.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

#### 5.9.1.7 Examples

- `match("test", "[0-9]+")` results in `false`
- `match("1234", "[0-9]+")` results in `true`
- `match("hello", "[a-z]+")` results in `true`

## 3.5.10 5.10 Is null (isnull)

#### 5.10.1.1 Syntax

**isnull(** op **)**

#### 5.10.1.2 Input Parameters

op: rset | scal <*>

#### 5.10.1.3 Output

rset | scal <boo>

#### 5.10.1.4 Semantics

Returns `true` if the value of op is `null`, `false` otherwise.

#### 5.10.1.5 Additional Constraints

None.

#### 5.10.1.6 Behaviour

- Binary *Operators'* standard behaviour.
- Comparison *Operators'* standard behaviour.

# 3.6 6 Logical Operators

Logical *Operators* describe operations involving two *Operands* with Boolean *Data Type*. To deal with null values, all operations implements Kleene logic for logical operations (also known as Three-valued logic).

## 3.6.1 6.1 Conjunction (and)

#### 6.1.1.1 Syntax

left **and** right

#### 6.1.1.2 Input Parameters

left: rset | scal <boo>

right: rset | scal <boo>

#### 6.1.1.3 Output

rset | scal

#### 6.1.1.4 Semantics

Returns `true` if both *Operands* are `true`, otherwise `false`.

#### 6.1.1.5 Additional Constraints

None.

#### 6.1.1.6 Behaviour

- Binary *Operators'* standard behaviour.

| | **false** | **null** | **true** |
|---|---|---|---|
| false | false | false | false |
| null | false | null | null |
| true | false | null | true |

## 3.6.2 6.2 Disjunction (or)

#### 6.2.1.1 Syntax

left **or** right

#### 6.2.1.2 Input Parameters

left: rset | scal <boo>

right: rset | scal <boo>

#### 6.2.1.3 Output

rset | scal <boo>

#### 6.2.1.4 Semantics

Returns `true` if any *Operand* is `true`, otherwise `false`.

#### 6.2.1.5 Additional Constraints

None.

#### 6.2.1.6 Behaviour

- Binary *Operators'* standard behaviour.

| | **false** | **null** | **true** |
|---|---|---|---|
| false | false | null | true |
| null | null | null | true |
| true | true | true | true |

## 3.6.3 6.3 Exclusive Disjunction (xor)

#### 6.3.1.1 Syntax

left **xor** right

#### 6.3.1.2 Input Parameters

left: rset | scal <boo>

right: rset | scal <boo>

#### 6.3.1.3 Output

rset | scal <boo>

#### 6.3.1.4 Semantics

Returns `true` if one *Operand* is `true` and the other is `false`, otherwise `false`.

#### 6.3.1.5 Additional Constraints

None.

#### 6.3.1.6 Behaviour

- Binary *Operators'* standard behaviour.

| | **false** | **null** | **true** |
|---|---|---|---|
| false | false | null | true |
| null | null | null | null |
| true | true | null | false |

## 3.6.4 6.4 Negation (not)

#### 6.4.1.1 Syntax

**not** op

#### 6.4.1.2 Input Parameters

op: rset | scal <boo>

#### 6.4.1.3 Output

rset | scal <boo>

#### 6.4.1.4 Semantics

Returns `true` if op is `false`, and `false` if op is `true`.

#### 6.4.1.5 Additional Constraints

None.

#### 6.4.1.6 Behaviour

- Binary *Operators'* standard behaviour.

| Input | Result |
|---|---|
| false | true |
| null | null |
| true | false |

# 3.7 7 Aggregate Operators

## 3.7.1 7.1 Aggregate Operators' General Behaviour

#### 7.1.1 Syntax

**aggregateOperator (** op {**group by** groupingId {, groupingId}*} **)**

#### 7.1.2 Input Parameters

op: rset <*>

groupingId: scal <prop>

#### 7.1.3 Output

rset | scal <*>

#### 7.1.4 Semantics

Aggregate *Operators* perform operations on the measures of the *Operand Recordset*, calculating the required aggregated values for groups of *Records*. The groups of *Records* to be aggregated are specified through the grouping clause. If no grouping clause is used, the operation shall be calculated on all the *Records*, resulting in a *Scalar*.

#### 7.1.5 Additional Constraints

The allowed *Data Types* depend on the specific *Operator* according to the following table:

| ***Operator*** | ***Operand* Type** | **Result Type** |
|---|---|---|
| Sum | Number | Number |
| Count | Any | Integer |
| Min | Any | Any |
| Max | Any | Any |
| Average | Number | Number |
| Median | Number | Number |

The *Components* in the grouping by clause shall be present in the *Operand*.

#### 7.1.6 Behaviour

Aggregate *Operations* generate a *Recordset* or a *Scalar*, depending on the grouping clause.

If the grouping clause exists, the structure of the resulting *Recordset* has as *Key Components* the *Components* in the group by

The result may be:

- A *Recordset*, for which the resulting structure contains as *Key* dimensions the *Components* included in the group by clause.
- A *Scalar*, if the grouping clause is omitted.

#### 7.1.7 Examples

Supposing the following *Recordset* with name rs1:

| r | c | CNT | f |
|---|---|---|---|
| 010 | 010 | PT | 100 |
| 010 | 020 | PT | 200 |
| 020 | 010 | PT | 300 |
| 020 | 020 | PT | 400 |
| 010 | 010 | DE | 500 |
| 010 | 020 | DE | 600 |
| 020 | 010 | DE | 700 |
| 020 | 020 | DE | 800 |

`sum(rs1 group by r)` results in:

| r | f |
|---|---|
| 010 | 1400 |
| 020 | 2200 |

`sum(rs1 group by r, CNT)` results in:

| r | CNT | f |
|---|---|---|
| 010 | PT | 300 |
| 020 | PT | 700 |
| 010 | DE | 1100 |
| 020 | DE | 1500 |

`count(rs1)` results in: `8` (*Scalar*)

## 3.7.2 7.2 Sum (sum)

#### 7.2.1 Syntax

**sum(** op {**group by** groupingId {, groupingId}*} **)**

#### 7.2.2 Input Parameters

op: rset <num + interval>

groupingId: scal <prop>

#### 7.2.3 Output

rset | scal <num>

#### 7.2.4 Semantics

Returns the sum of the input values. Follows the general semantics of aggregate *Operators*.

For intervals:

- The centre is calculated as the sum of the all the centers of the *Operands*.
- The radius is calculated as the sum of all the radiuses of the *Operands*.

#### 7.2.5 Additional Constraints

None.

#### 7.2.6 Behaviour

Aggregate *Operators'* general behaviour.

### 3.7.3 7.3 Count (count)

#### 7.3.1 Syntax

**count(** op {**group by** groupingId {, groupingId}*} **)**

#### 7.3.2 Input Parameters

op: rset <*>

groupingId: scal <prop>

#### 7.3.3 Output

rset | scal <num>

#### 7.3.4 Semantics

Returns the number of *Records* in the *Recordset* or groups of *Records*. Follows the general semantics of aggregate *Operators*.

#### 7.3.5 Additional Constraints

None.

#### 7.3.6 Behaviour

Aggregate *Operators'* general behaviour.

Note: Aggregate *Operators* generally ignore null values. This behavior can be overridden by using the nvl *Operator*.

### 3.7.4 7.4 Minimum Value (min_aggr)

#### 7.4.1 Syntax

**min_aggr(** op {**group by** groupingId {, groupingId}*} **)**

#### 7.4.2 Input Parameters

op: rset <*>

groupingId: scal <prop>

#### 7.4.3 Output

rset | scal <num>

#### 7.4.4 Semantics

Returns the minimum value of the input values. Follows the general semantics of aggregate *Operators*.

For intervals:

- The centre is calculated as the minimum value of the all the centers of the *Operands*.
- The radius is the radius of the *Operand* that has the minimum centre.

#### 7.4.5 Additional Constraints

None.

#### 7.4.6 Behaviour

Aggregate *Operators'* general behaviour.

### 3.7.5 7.5 Maximum Value (max_aggr)

#### 7.5.1 Syntax

**max_aggr(** op {**group by** groupingId {, groupingId}*} **)**

#### 7.5.2 Input Parameters

op: rset <*>

groupingId: scal <prop>

#### 7.5.3 Output

rset | scal <num>

#### 7.5.4 Semantics

Returns the maximum value of the input values. Follows the general semantics of aggregate *Operators*.

For intervals: - The centre is calculated as the maximum value of the all the centers of the *Operands*. - The radius is the radius of the *Operand* that has the maximum centre.

#### 7.5.5 Additional Constraints

None.

#### 7.5.6 Behaviour

Aggregate *Operators'* general behaviour.

### 3.7.6 7.6 Average (avg)

#### 7.6.1 Syntax

**avg(** op {**group by** groupingId {, groupingId}*} **)**

#### 7.6.2 Input Parameters

op: rset <num>

groupingId: scal <prop>

#### 7.6.3 Output

rset | scal <num>

#### 7.6.4 Semantics

Returns the average of the input values. Follows the general semantics of aggregate *Operators*.

#### 7.6.5 Additional Constraints

None.

#### 7.6.6 Behaviour

Aggregate *Operators'* general behaviour.

## 3.7.7 7.7 Median Value (median)

#### 7.7.1 Syntax

**median(** op {**group by** groupingId {, groupingId}*} **)**

#### 7.7.2 Input Parameters

op: rset <num>

groupingId: scal <prop>

#### 7.7.3 Output

rset | scal <num>

#### 7.7.4 Semantics

Returns the median of the input values. Follows the general semantics of aggregate *Operators*.

#### 7.7.5 Additional Constraints

None.

#### 7.7.6 Behaviour

Aggregate *Operators'* general behaviour.

# 3.8 8 Conditional Operators

## 3.8.1 8.1 If-then-else

#### 8.1.1 Syntax

**if** conditionExpression **then** thenExpression {**else** elseExpression} **endif**

#### 8.1.2 Input Parameters

conditionExpression: rset | scal <boo>

thenExpression: rset | scal <*>

elseExpression: rset | scal <*>

#### 8.1.3 Output

rset | scal <*>

#### 8.1.4 Semantics

Returns the thenExpression if the conditionExpression evaluates to `true`, elseExpression otherwise.

#### 8.1.5 Additional Constraints

The thenExpression and elseExpression must be of the same *Data Type* or a compatible one. In case they are not of the same *Data Type*, there shall be implicit casting to the common *Data Type*, which will be the *Data Type* of the result

The thenExpression and elseExpression need to have the same data structure.

If the conditionExpression is of *Recordset* type, the data structures of the thenExpression and elseExpression must contain the same identifiers as the conditionExpression, or a subset of them. The resulting structure will contain the *Components* of the conditionExpression *Recordset*.

The following table contains the applicability options for the if-then-else *Operator*:

| Condition | Then | Else | Is allowed | Result structure | Remarks |
|---|---|---|---|---|---|
| *Scalar* | *Scalar* | None | Yes | *Scalar* | |
| *Scalar* | *Recordset* | None | No | - | Forbidden because the result structure would be unknown |
| *Scalar* | *Scalar* | *Scalar* | Yes | *Scalar* | |
| *Scalar* | *Scalar* | *Recordset* | No | - | Forbidden because the result structure would be unknown |
| *Scalar* | *Recordset* | *Scalar* | No | - | Forbidden because the result structure would be unknown |
| *Scalar* | *Recordset* | *Recordset* | Yes | *Recordset* | |
| *Recordset* | *Scalar* | None | Yes | *Recordset* | |
| *Recordset* | *Recordset* | None | Yes | *Recordset* | |
| *Recordset* | *Scalar* | *Scalar* | Yes | *Recordset* | |
| *Recordset* | *Scalar* | *Recordset* | Yes | *Recordset* | |
| *Recordset* | *Recordset* | *Scalar* | Yes | *Recordset* | |
| *Recordset* | *Recordset* | *Recordset* | Yes | *Recordset* | |

#### 8.1.6 Behaviour

For conditionExpressions of the *Scalar* type, returns the *Operand* resulting from the thenExpression or the elseExpression.

For conditionExpressions of the *Recordset* type, returns a *Recordset* with the data structure of the conditionExpression, with one output *Record* per input *Record*. The *Fact* values will be the corresponding for the thenExpression or elseExpression, depending on the evaluation result of the condition.

If the elseExpression is omitted, a null is returned, except for the case when the thenExpression is of the Boolean *Data Type*, in which case true is returned.

If the condition evaluates to null, the elseExpression is returned.

#### 8.1.7 Examples

## 3.8.2 8.2 Null Substitute (nvl)

#### 8.2.1 Syntax

**nvl(** op1, op2 **)**

#### 8.2.2 Input Parameters

op1: rset | scal <*>

op2: rset | scal <*>

#### 8.2.3 Output

rset | scal <*>

#### 8.2.4 Semantics

Returns op2 when op1 is `null`, otherwise op1.

#### 8.2.5 Additional Constraints

op1 and op2 need to be of the same type (*Scalar* or *Recordset*) and same *Data Type*.

If op1 and op2 are of the *Recordset* type, they need to have the same data structure.

#### 8.2.6 Behaviour

Binary *Operators'* standard behaviour.

#### 8.2.7 Examples

## 3.8.3 8.3 Filter

#### 8.3.1 Syntax

**filter(** filteredOp, filteringOp **)**

#### 8.3.2 Input Parameters

filteredOp: rset <*>

filteringOp: rset <bool>

#### 8.3.3 Output

rset <*>

#### 8.3.4 Semantics

Returns the *Records* of the filteredOp *Recordset* for which their matching *Record* in filteringOp evaluates to true.

#### 8.3.5 Additional Constraints

filteringOp must have the same *Key Components* as filteredOp or a subset of them.

#### 8.3.6 Behaviour

Returns a *Recordset* with the same data structure as filteredOp.

An inner join between filteredOp and filteringOp is performed. The *Records* in filteredOp that match to a *Record* in filteringOp with true value are returned.

#### 8.3.7 Examples

# 3.9 9 String Operators

## 3.9.1 9.1 Length (len)

#### 9.1.1 Syntax

**len(** op **)**

#### 9.1.2 Input Parameters

op: rset | scal <str>

#### 9.1.3 Output

rset | scal <int>

#### 9.1.4 Semantics

Returns the number of characters of the op string.

#### 9.1.5 Additional Constraints

None.

#### 9.1.6 Behaviour

- Unary *Operators'* standard behaviour.

#### 9.1.7 Examples

- `len("test")` results in `4`
- `len(null)` results in `null`

## 3.9.2 9.2 Concatenate (&)

#### 9.2.1 Syntax

op1 **&** op2

#### 9.2.2 Input Parameters

op1: rset | scal <str>

op2: rset | scal <str>

#### 9.2.3 Output

rset | scal <str>

#### 9.2.4 Semantics

Concatenates two strings.

#### 9.2.5 Additional Constraints

None.

#### 9.2.6 Behaviour

- Binary *Operators'* standard behaviour.

#### 9.2.7 Examples

- `"hello" & "world"` results in `"helloworld"`
- `"test" & null` results in `null`

# 3.10 10 Time Operators

## 3.10.1 10.1 Time Shift

#### 10.1.1 Syntax

**time_shift(** op, period, numberPeriods, {var} **)**

#### 10.1.2 Input Parameters

op: rset <*> | scal <date>

period: scal <str>

numberPeriods: scal <int>

var: scal <prop>

#### 10.1.3 Output

rset <*>

#### 10.1.4 Semantics

Changes the dates of the var *Component* of the *Recordset* op by adding (or subtracting) the numberPeriods of the period type.

#### 10.1.5 Additional Constraints

- If op is a *Recordset*, then
- var must exist,
- must belong to the *Recordset* op, and
- must be of Time Interval type.
- If var is an *Scalar*, then var cannot exist.

The period must have one of the following values:

- *A* for year
- *S* for semester
- *Q* for quarter
- *M* for month
- *W* for week
- *D* for day

#### 10.1.6 Behaviour

Returns a *Recordset* with the same data structure and number of *Records* as the input op. The dates for the *Component* var are modified by adding the numberPeriods (subtracting, if negative) of the period type.

#### 10.1.7 Examples

Considering the following *Recordset*, generated from the selection {tT1, r010-020, c010-020}:

| RefDate | r | c | f |
|---|---|---|---|
| 2022Q1 | 010 | 010 | 100 |
| 2022Q1 | 010 | 020 | 200 |
| 2022Q1 | 010 | 010 | 300 |
| 2022Q1 | 010 | 020 | 400 |
| 2022Q2 | 020 | 010 | 500 |
| 2022Q2 | 020 | 020 | 600 |
| 2022Q2 | 020 | 010 | 700 |
| 2022Q2 | 020 | 020 | 800 |

`time_shift({tT1, r010-020, c010-020}, Q, 1, refPeriod)` returns:

| RefDate | R | c | f |
|---|---|---|---|
| 2022Q2 | 010 | 010 | 100 |
| 2022Q2 | 010 | 020 | 200 |
| 2022Q2 | 010 | 010 | 300 |
| 2022Q2 | 010 | 020 | 400 |
| 2022Q3 | 020 | 010 | 500 |
| 2022Q3 | 020 | 020 | 600 |
| 2022Q3 | 020 | 010 | 700 |
| 2022Q3 | 020 | 020 | 800 |

`time_shift({tT1, r010-020, c010-020}, Q, -1, refPeriod)` returns:

| RefDate | r | c | f |
|---|---|---|---|
| 2021Q4 | 010 | 010 | 100 |
| 2021Q4 | 010 | 020 | 200 |
| 2021Q4 | 010 | 010 | 300 |
| 2021Q4 | 010 | 020 | 400 |
| 2022Q1 | 020 | 010 | 500 |
| 2022Q1 | 020 | 020 | 600 |
| 2022Q1 | 020 | 010 | 700 |
| 2022Q1 | 020 | 020 | 800 |

# 3.11 11 Clause Operators

Clause *Operators* serve to perform operations on the DPM *Key Components* of *Recordsets*. Use of clause *Operators* with Standard *Key Components* are not allowed.

## 3.11.1 11.1 where

#### 11.1.1 Syntax

op **[where** condition **]**

#### 11.1.2 Input Parameters

op: rset <*>

condition: expression <boo>

#### 11.1.3 Output

rset <*>

#### 11.1.4 Semantics

Filters a *Recordset* based on the value of a *Key Component*.

#### 11.1.5 Additional Constraints

Condition must be a Boolean expression using as input *Key Components* of op.

#### 11.1.6 Behaviour

Returns a *Recordset* with the same data structure as the input *Operand*, and with the *Records* resulting from the evaluation of the filtering condition. When the condition evaluates to true, the *Record* is kept, otherwise (including null), the *Record* is not kept.

#### 11.1.7 Examples

Considering the following *Recordset*, generated from the selection `{tT1, r010-020}`:

| RefDate | r | CNT | f |
|---|---|---|---|
| 2022Q1 | 010 | ES | 100 |
| 2022Q1 | 010 | PT | 200 |
| 2022Q1 | 010 | DE | 300 |
| 2022Q1 | 010 | IT | 400 |
| 2022Q2 | 020 | ES | 500 |
| 2022Q2 | 020 | PT | 600 |
| 2022Q2 | 020 | DE | 700 |
| 2022Q2 | 020 | | 800 |

`{tT1, r010-020}[where CNT in {"ES", "PT"}]` returns:

| RefDate | r | CNT | f |
|---|---|---|---|
| 2022Q1 | 010 | ES | 100 |
| 2022Q1 | 010 | PT | 200 |
| 2022Q2 | 020 | ES | 500 |
| 2022Q2 | 020 | PT | 600 |

## 3.11.2 11.2 rename

#### 11.2.1 Syntax

op **[rename** compFrom to compTo **]**

#### 11.2.2 Input Parameters

op: rset <*>

compFrom: scal <prop>

compTo: scal <prop>

#### 11.2.3 Output

rset <*>

#### 11.2.4 Semantics

Changes the name of a *Component*.

#### 11.2.5 Additional Constraints

compFrom must belong to op.

compFrom must be a DPM *Key Component* (i.e., cannot refer to row, column or sheet).

compTo name cannot be already used in op.

compTo name cannot be the name of a standard *Key Component* ("r", "c" or "s").

#### 11.2.6 Behaviour

Returns a *Recordset* with the same data structure as the input *Operand*, except the renamed *Components*, which change name. All the *Records* are kept.

#### 11.2.7 Examples

Considering the following *Recordset*, generated from the selection `{tT1, r010-020}` :

| RefDate | r | CNT | f |
|---|---|---|---|
| 2022Q1 | 010 | ES | 100 |
| 2022Q1 | 010 | PT | 200 |
| 2022Q1 | 010 | DE | 300 |
| 2022Q1 | 010 | IT | 400 |
| 2022Q2 | 020 | ES | 500 |
| 2022Q2 | 020 | PT | 600 |
| 2022Q2 | 020 | DE | 700 |
| 2022Q2 | 020 | | 800 |

`{tT1, r010-020}[rename CNT to country]` returns:

| RefDate | r | country | f |
|---|---|---|---|
| 2022Q1 | 010 | ES | 100 |
| 2022Q1 | 010 | PT | 200 |
| 2022Q1 | 010 | DE | 300 |
| 2022Q1 | 010 | IT | 400 |
| 2022Q2 | 020 | ES | 500 |
| 2022Q2 | 020 | PT | 600 |
| 2022Q2 | 020 | DE | 700 |
| 2022Q2 | 020 | | 800 |

## 3.11.3 11.3 get

#### 11.3.1 Syntax

op **[get** component **]**

#### 11.3.2 Input Parameters

op: rset <*>

component: scal <prop>

#### 11.3.3 Output

rset <*>

#### 11.3.4 Semantics

Returns a *Recordset* where the fact is the value of one of the *Key Components*.

#### 11.3.5 Additional Constraints

component must belong to op.

#### 11.3.6 Behaviour

Returns a *Recordset* with the same data structure as the input *Operand*. The *Fact* values the *Records* are substituted for the value of the selected *Component*.

*Records* with null *Fact* value will be omitted.

#### 11.3.7 Examples

Considering the following *Recordset*, generated from the selection `{tT1, r010-020}`:

| RefDate | r | CNT | f |
|---|---|---|---|
| 2022Q1 | 010 | ES | 100 |
| 2022Q1 | 010 | PT | 200 |
| 2022Q1 | 010 | DE | 300 |
| 2022Q1 | 010 | IT | 400 |
| 2022Q2 | 020 | ES | 500 |
| 2022Q2 | 020 | PT | 600 |
| 2022Q2 | 020 | DE | 700 |
| 2022Q2 | 020 | IT | 800 |

`{tT1, r010-020}[get CNT]` returns:

| RefDate | r | CNT | f |
|---|---|---|---|
| 2022Q1 | 010 | ES | ES |
| 2022Q1 | 010 | PT | PT |
| 2022Q1 | 010 | DE | DE |
| 2022Q1 | 010 | IT | IT |
| 2022Q2 | 020 | ES | ES |
| 2022Q2 | 020 | PT | PT |
| 2022Q2 | 020 | DE | DE |
| 2022Q2 | 020 | IT | IT |

# 4. Grammar

## 4.1 Introduction

The DPM-XL grammar provides the technical definition of the syntax of the language. It allows developers to build parsers for the language. It is provided in two .g4 files. The grammar usses the Extended Backus-Naur Form (EBNF)

Grammar files:

- Lexer
- parser